

B 题：更快的快通系统

无论是在收费站、游乐场或其他地方正出现着越来越多的“快通”系统以减少人们排队等候的时间。请考虑一家游乐场的快通系统的设计。这家游乐场已经为几种受欢迎的乘骑项目提供快通系统的服务作为试验。该系统的设计思想是对某些受欢迎的乘骑项目，游客可以到该娱乐项目旁边的一个机器前并将当天的门票插入，该机器将返回给你一张纸条，上面写着你可以在某个特定的时间段回来。比如说你把你的门票在 1:15pm 插到机器里，快通系统就告诉你可以在 3:30—4:30pm 回来，你可以凭你的纸条第二次排队，这时队伍可能比较短，你就可以较快进入景点。为了防止游客同时在几个乘骑娱乐项目上使用这个系统。一个顾客在同一时刻只能得到一次快通系统的服务。

为改进快通系统的运作你们队被聘为几个合格的顾问之一。游客一直在抱怨该试验系统的一些异常现象。比如说，顾客有时看到快通系统提供的回到景点时间是 4 小时以后。但是才过一小会，在相同的景点系统所提供的回到景点的时间只有 1 小时或稍多一点时间。有时按照快通系统安排的游客的人数和等待时间几乎和正常排队的人数和所花费的时间一样多。

于是问题就是要提出并检验能提高快通系统效率的方案以使人们可以更多地享受在游乐场的休闲时光。问题的一部分是要确定评估各种可供选择的方案的评价准则。你们的报告中要包括一份非技术性的概述，以便游乐场主管从各个顾问所提出的可供选择的方案中作出选择。

PROBLEM B: A Faster QuickPass System

“QuickPass” systems are increasingly appearing to reduce people’s time waiting in line, whether it is at tollbooths, amusement parks, or elsewhere. Consider the design of a QuickPass system for an amusement park. The amusement park has experimented by offering QuickPasses for several popular rides as a test. The idea is that for certain popular rides you can go to a kiosk near that ride and insert your daily park entrance ticket, and out will come a slip that states that you can return to that ride at a specific time later. For example, you insert your daily park entrance ticket at 1:15 pm, and the QuickPass states that you can come back between 3:30 and 4:30 pm when you can use your slip to enter a second, and presumably much shorter, line that will get you to the ride faster. To prevent people from obtaining QuickPasses for several rides at once, the QuickPass machines allow you to have only one active QuickPass at a time. You have been hired as one of several competing consultants to improve the operation of QuickPass. Customers have been complaining about some anomalies in the test system. For example, customers observed that in one instance QuickPasses were being offered for a return time as long as 4 hours later. A short time later on the same ride, the QuickPasses were given for times only an hour or so later. In some instances, the lines for people with Quickpasses are nearly as long and slow as the regular lines.

The problem then is to propose and test schemes for issuing QuickPasses

in order to increase people's enjoyment of the amusement park. Part of the problem is to determine what criteria to use in evaluating alternative schemes. Include in your report a non-technical summary for amusement park executives who must choose between alternatives from competing consultants.